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Electrical engineering degree, curriculum: Electronics for intelligent systems, big-data and internet of things

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Analysis and circuital implementation of Discrete Fourier Transform on Phase-change in-Memory Computing technology

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Chapter 1

Introduction

Modern embedded and low-resource applications demand high-speed signal processing under strict energy and hardware constraints. While frequency-domain analysis via the Discrete Fourier Transform (DFT) and Fast Fourier Transform (FFT) provides essential spectral insights, traditional Von Neumann architectures suffer from severe bottlenecks due to the high energy and latency costs of moving data between separated memory and processing units.

To overcome this, Analog in-Memory Computing (AiMC) emerges as a promising non-Von Neumann paradigm. By leveraging Phase-Change Memory (PCM) technology, linear matrix-vector multiplications—such as those required by the DFT and FFT—can be executed directly within non-volatile memory arrays using analog conductance values.

However, hardware implementation introduces challenges such as device variability, noise, and conductance drift over time. In this work are presented other algorithms to calculate the DFT in real time focusing on different types of algorithmic errors and PCM device non-idealities, and a small circuital implementation of one algorithm.

Chapter 2

Algorithms

2.1 DFT and then FFT

The Discrete Fourier Transform (DFT) maps a finite sequence of N_0 samples, $x[n]$, to its frequency-domain representation, X_k. Using the forward-transform convention adopted here, the DFT and its inverse are

$$X_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] W_{N_0}^{kn},$$
$$x[n] = \sum_{k=0}^{N_0-1} X_k W_{N_0}^{-kn},$$

where $n, k \in \{0, 1, \ldots, N_0 - 1\}$ and

$$W_{N_0} = e^{-j2\pi/N_0}$$

is the primitive N_0-th root of unity. Its powers, $W_{N_0}^{kn}$, are called twiddle factors.

A direct evaluation of the DFT requires N_0^2 complex multiplications and $N_0(N_0 - 1)$ complex additions. Counting a general complex multiplication as four real multiplications and two real additions, and a complex addition as two real additions, this corresponds to $4N_0^2$ real multiplications and $4N_0^2 - 2N_0$ real additions. This count assumes that no operations are eliminated for special twiddle factors. Therefore, the direct DFT has a computational complexity of $O(N_0^2)$.

The Fast Fourier Transform (FFT) computes the same DFT values without approximation while reducing the number of operations. The two common radix-2 forms are decimation in time (DIT) and decimation in frequency (DIF). The derivation below uses the DIT form and assumes that N_0 is a power of two.

In the first DIT decomposition, the input sequence is separated into its even- and odd-indexed samples:

$$e[n] = x[2n], \qquad o[n] = x[2n + 1], \qquad 0 \le n < \frac{N_0}{2}.$$

Let E_k and O_k denote the $N_0/2$-point DFTs of $e[n]$ and $o[n]$, respectively. For $0 \le k <$

$N_0/2$, splitting the DFT sum into its even- and odd-indexed terms gives

$$\begin{aligned} X_k &= \sum_{n=0}^{N_0/2-1} x[2n] W_{N_0}^{2kn} + \sum_{n=0}^{N_0/2-1} x[2n+1] W_{N_0}^{k(2n+1)} \\ &= \sum_{n=0}^{N_0/2-1} e[n] W_{N_0/2}^{kn} + W_{N_0}^{k} \sum_{n=0}^{N_0/2-1} o[n] W_{N_0/2}^{kn} \\ &= E_k + W_{N_0}^{k} O_k. \end{aligned}$$

The other half of the output follows from $W_{N_0}^{N_0/2} = -1$:

$$X_{k+N_0/2} = E_k - W_{N_0}^{k} O_k, \qquad 0 \le k < \frac{N_0}{2}.$$

Together, these two equations define a butterfly operation showed in Fig.2.1: for inputs E_k and O_k, the outputs are $E_k + W_{N_0}^{k} O_k$ and $E_k - W_{N_0}^{k} O_k$.

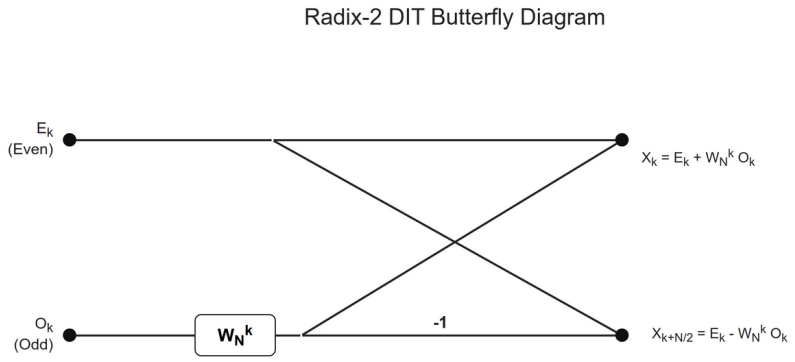


Figure 2.1: Radix-2 DIT butterfly diagram

For $N_0 = 2^m$, a radix-2 FFT has $m = \log_2 N_0$ stages, each containing $N_0/2$ butterflies. Counting one twiddle-factor multiplication, including trivial factors, and two complex additions per butterfly, the algorithm requires $(N_0/2)\log_2 N_0$ twiddle-factor multiplications and $N_0 \log_2 N_0$ complex additions. Its computational complexity is therefore $O(N_0 \log_2 N_0)$.

2.2 Sliding DFT

The FFT algorithm requires the entire signal to calculate the spectrum. However, some applications (such as tuning vibration neutralizers) require a faster algorithm that operates in real time. The article [3] presents two recursive algorithms, both referred to as the sliding DFT (SDFT). The core idea is to compute the initial spectrum using the first N_0 samples and then update it recursively whenever a new sample arrives, dropping the oldest one. In Fig.2.2 is showed an example of a sliding window.

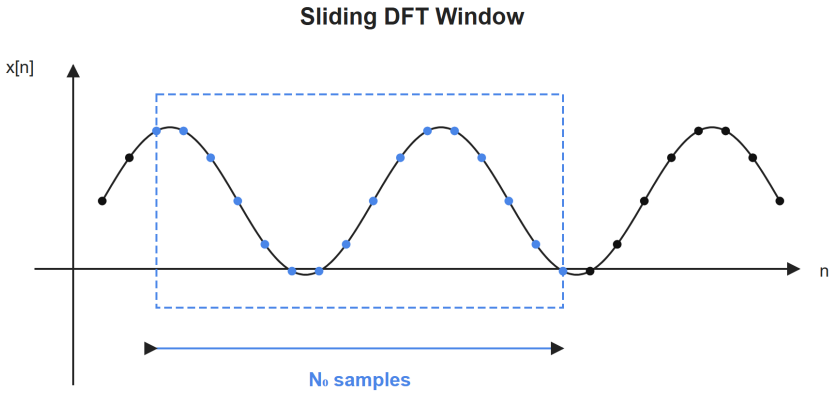


Figure 2.2: Enter Caption

2.3 R-SDFT and M-SDFT

In the article [3] are studied 2 algorithms: resonator and modulator used in branches included in a bigger recursive structure showed in the figure Fig. 2.3. The transfer function of the kth branch of the SDFT using a general forumlation is:

$$H_{SDFT}(z) = \frac{X_k(z)}{X(z)} = \frac{H_k(z)}{1 + \frac{1}{N}\sum_{k=0}^{N-1} H_k(z)}$$

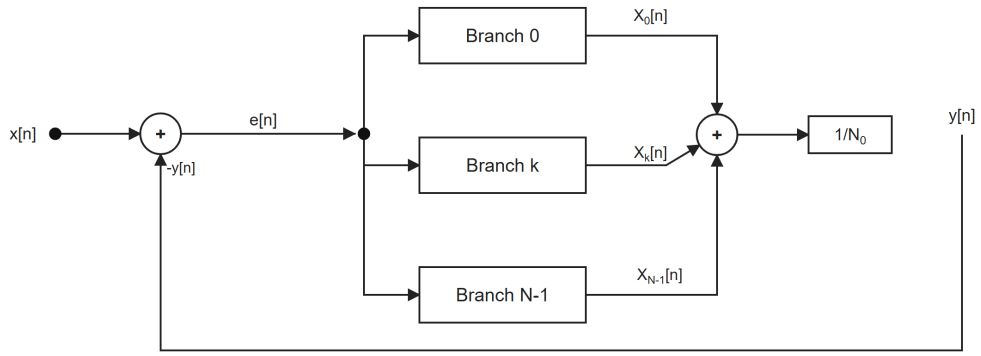


Figure 2.3: Resonator-based sliding DFT branch

2.3.1 R-SDFT

In the Fig.2.4 is showed the resonator branch. In the following the calculations to find the relation of this algorithm.

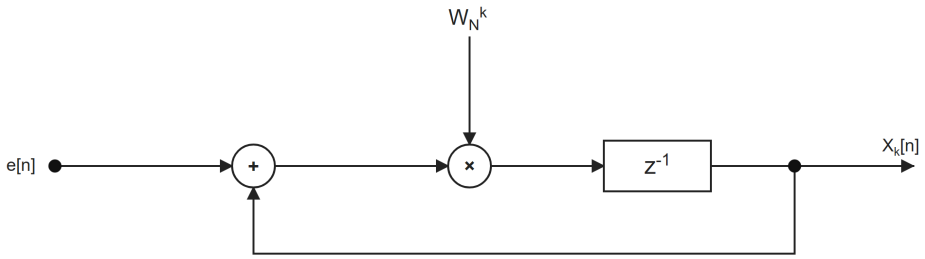


Figure 2.4: Resonator-based sliding DFT branch

$$\begin{aligned} t[n] &= r[n] \cdot W_{N_0}^k = (e[n] + X_k[n]) \cdot W_{N_0}^k \\ X_k[n] &= t[n-1] = (e[n-1] + X_k[n-1]) \cdot W_{N_0}^k \end{aligned} \xrightarrow{\text{Z-transform}} \begin{aligned} X_k(z) &= (E(z)z^{-1} + X_k(z)z^{-1}) \cdot W_N^k \\ X_k(z) &= E(z) \cdot \frac{W_N^k z^{-1}}{1 - W_N^k z^{-1}} \end{aligned}$$

This structure has a pole in W_N^k for this reason the structure is more affected by the quantization error with respect to the modulator-SDFT.
In the Listing 2.1 is showed the resonator-SDFT.

```
def rsdft(input_data:None, N:10, sample_rate:int=1, quant_en:bool=True, num_bits:int
    =8, max_amplitude:float=1.0,
          r_ideal:float=1.0, r_quant:float=1.0, hw_err:bool=False):

    input_data = input_data[:N]
    input_size = input_data.size

    k_vec = np.arange(N)
    if hw_err:
        rng = np.random.default_rng()
        k_vec = k_vec + rng.uniform(low=LOW_ERR, high=HIGH_ERR, size=N)
    arg = 2 * np.pi * k_vec / N

    # check of the constant
    v = [1.0]
    if SEARCH_R and ((quant_en and r_quant in v) or (not quant_en and r_ideal in v)):
        ok = False
        while(ok==False):
            if quant_en:
                twiddles_quant = quantizer(r_quant * np.exp(-1j * arg), num_bits=
                    num_bits)
                if np.all(np.abs(twiddles_quant) <= 1.0): ok = True
                else: r_quant -= 0.001
            else:
                twiddles_ideal = r_ideal * np.exp(-1j * arg)
                if np.all(np.abs(twiddles_ideal) <= 1.0): ok = True
                else: r_ideal -= 0.001
    else:
```

```
        if quant_en:    twiddles_quant = quantizer(r_quant * np.exp(-1j * arg),
            num_bits=num_bits)
        else:           twiddles_ideal = r_ideal * np.exp(-1j * arg)

    # variables
    out = np.zeros(input_size, dtype=np.float64)
    Z = np.zeros(N, dtype=np.complex128)
    prev_out = 0.0
    err = 0.0

    for index in range(input_size):
        if quant_en:
            err = quantizer(input_data[index] - prev_out, num_bits=num_bits)
            Z = quantizer(quantizer((err + Z), num_bits=num_bits) * twiddles_quant,
                num_bits=num_bits)
            prev_out = quantizer(np.sum(np.real(Z)) / N, num_bits=num_bits)
        else:
            err = input_data[index] - prev_out
            Z = (err + Z) * twiddles_ideal
            prev_out = np.sum(np.real(Z)) / N

        out[index] = prev_out

    return Z, r_ideal, r_quant
```

Listing 2.1: Resonator-based sliding DFT Python code

2.3.2 M-SDFT

In the Fig.2.5 is showed the modulator branch. In the following the calculations to find the relation of this algorithm. This algorithm can be divided in 3 parts: modulation, integration and demodulation. The modulation and the demodulation parts are defined by the multiplication of $W_{N_0}^{kn}$ and $W_{N_0}^{-kn}$, instead the integrator has the following transfer function.

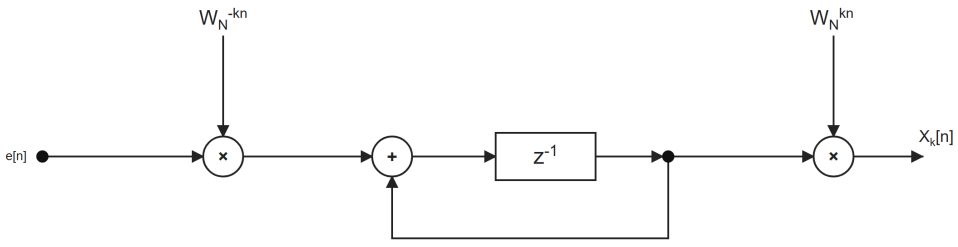


Figure 2.5: Modulator-based sliding DFT branch

$$\begin{aligned} u[n] &= e[n] \cdot W_{N_0}^{-kn} \\ v[n] &= u[n] + v[n-1] \end{aligned} \xrightarrow{\text{Z-transform}} \begin{aligned} V(z) &= U(z) + V(z)z^{-1} \\ \frac{V(z)}{U(z)} &= \frac{1}{1-z^{-1}} \end{aligned}$$

The pole in $|z| = 1$ ensures more stability than the resonator one from the fact is less affected by the quantization algorithm.
In the Listing 2.2 is showed the modulator-based SDFT Python code.

```python
def msdft(input_data:None, N:10, sample_rate:int=1, quant_en:bool=True, num_bits:int=8, max_amplitude:float=1.0,
          r_ideal:float=1.0, r_quant:float=1.0, hw_err:bool=False):

    input_data = input_data[:N]
    input_size = input_data.size

    k_vec = np.arange(N)
    if hw_err:
        rng = np.random.default_rng()
        k_vec = k_vec + rng.uniform(low=LOW_ERR, high=HIGH_ERR, size=N)
    arg = 2 * np.pi * k_vec / N

    # check of the constant
    v = [1.0]
    if SEARCH_R and ((quant_en and r_quant in v) or (not quant_en and r_ideal in v)):
        ok = False
        while(ok==False):
            if quant_en:
                for n in range(input_size):
                    twiddles_quant = quantizer(r_quant * np.exp(-1j * arg * n), num_bits=num_bits)
                    if not np.all(np.abs(twiddles_quant) <= 1.0):
                        r_quant -= 0.001
                        break
                    else:   ok = True
            else:
                for n in range(input_size):
                    twiddles_ideal = r_ideal * np.exp(-1j * arg * n)
                    if not np.all(np.abs(twiddles_ideal) <= 1.0):
                        r_ideal -= 0.001
                        break
                    else:   ok = True

    # variables
    out = np.zeros(input_size, dtype=np.float64)
    Z = np.zeros(N, dtype=np.complex128)
    prev_out = 0.0
    err = 0.0

    for index in range(input_size):
        if quant_en:
            err = quantizer(input_data[index] - prev_out, num_bits=num_bits)
            twiddle = quantizer(r_quant * np.exp(-1j * arg * index), num_bits=num_bits)
            prev_out = quantizer(np.sum(np.real(Z * np.conjugate(twiddle))) / N, num_bits=num_bits)
            Z = quantizer(quantizer(r_quant * Z, num_bits=num_bits) +
                    quantizer(err * twiddle, num_bits=num_bits), num_bits=num_bits)
        else:
            err = input_data[index] - prev_out
            twiddle = r_ideal * np.exp(-1j * arg * index)
            prev_out = np.sum(np.real(Z * np.conjugate(twiddle))) / N
            Z = r_ideal * Z + err * twiddle

        out[index] = prev_out

    return Z, r_ideal, r_quant
```

Listing 2.2: Modulator-based sliding DFT Python code

2.4 DC considerations

Working on the progect has been discovered a way to reduce energy consumption in the optic of an hardware implementation and is applicable to the general structure of Fig.2.3. Consist on considering only the DC component during the construction of the input signal.
From the definition of the spectrum components $X_0 = \sum_{n=0}^{N_0-1} x[n]$ so the reconstructed output is: $y[n] = x[n]$. This means there is no need of considering all other elements of the spectrum because the error is memorized from the previous data sample.

2.4.1 Resonator-based SDFT considering the DC component

In Listing 2.3 is showed the algorithm of the R-SDFT considering the DC component in the signal reconstruction. All the rest of the function is equal to the R-SDFT function.

```
for index in range(input_size):
    if quant_en:
        err = quantizer(input_data[index] - prev_out, num_bits=num_bits)
        Z = quantizer(quantizer((err + Z), num_bits=num_bits) * twiddles_quant,
            num_bits=num_bits)
        prev_out = quantizer(np.sum(np.real(Z[0])) / N, num_bits=num_bits)
    else:
        err = input_data[index] - prev_out
        Z = (err + Z) * twiddles_ideal
        prev_out = np.sum(np.real(Z[0])) / N

    out[index] = prev_out
```

Listing 2.3: R-SDFT DC component Python code part

2.4.2 Modulator-based SDFT considering the DC component

In Listing 2.4 is showed the algorithm of the M-SDFT considering the DC component in the signal reconstruction. All the rest of the function is equal to the M-SDFT function.

```
for index in range(input_size):
    if quant_en:
        err = quantizer(input_data[index] - prev_out, num_bits=num_bits)
        twiddle = quantizer(r_quant * np.exp(-1j * arg * index), num_bits=num_bits)
        prev_out = quantizer(np.sum(np.real(Z[0])) / N, num_bits=num_bits)
        Z = quantizer(quantizer(r_quant * Z, num_bits=num_bits) +
                quantizer(err * twiddle, num_bits=num_bits), num_bits=num_bits)
    else:
        err = input_data[index] - prev_out
        twiddle = r_ideal * np.exp(-1j * arg * index)
        prev_out = np.sum(np.real(Z[0])) / N
        Z = r_ideal * Z + err * twiddle

    out[index] = prev_out
```

Listing 2.4: M-SDFT DC component Python code part

Chapter 3

Results

This chapter presents the results from the Python code runs for different cases. The test bench consist of 200 waves using 3 variables parameters listed below:

- amplitude: ten amplitudes normalized to 1 $(0.1, 0.2, ..., 1.0)$;
- frequency: ten frequencies in the range from the first on-bin frequency to 15kHz (are excluded DC and Nyquist components to simplify the comparison);
- on-bin and off-bin: useful to test the algorithm with frequencies equal to the bins in the spectrum and other frequencies slightly different to the bins of the spectrum.

Each wave is defined as

$$\begin{aligned} x(t) &= A\sin(2\pi f_{sw} t), \\ f_{sw} &= f_{bin}(1-d), \\ f_{bin} &= \frac{f_s}{N_0}. \end{aligned}$$

The N_0 time samples are distributed linearly over the interval from 0 to N_0/f_s. where A is the amplitude of the wave, f_s is the sampling frequency, N_0 is the number of wave samples and of frequency bins and d is the off-bin factor (set to 0.25 in this analysis).

Subsequently, 3 different cases are analyzed:

- ideal spectrum used for comparison with next steps;
- 32, 24, 16, 12 and 8 bits quantization cases;
- the previous point adding the consideration of the PCM error conductance.

To evaluate the behaviour of these algorithms under these conditions the follwing metrics are used:

- magnitude error vs frequency

$$E_{mag} = \frac{\| |X_{algorithm}| - |X_{FFT}| \|_2}{\| |X_{FFT}| \|_2}$$

- complex (phase) error vs frequency

$$E_{complex} = \frac{\| X_{algorithm} - X_{FFT} \|_2}{\| X_{FFT} \|_2}$$

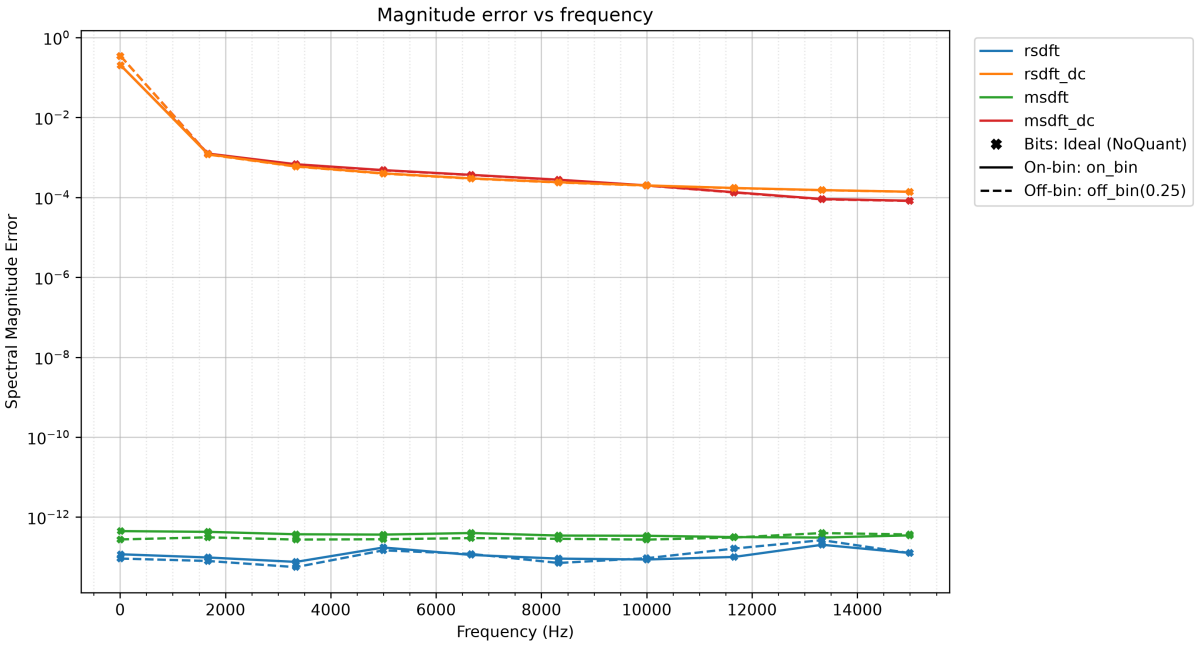


Figure 3.1: Ideal results case analysis, magnitude error vs frequency. $r = 1$ for every algorithm

- amplitude vs frequency vs complex (phase) error
- amplitude vs frequency vs magnitude error

Initially, a constant factor of $r = 1$ is assumed, followed by results obtained using the optimized r value for each algorithm.

3.1 Ideal (float64)

To perform this ideal analysis, the float64 precision offered by NumPy [2] played a fundamental role. T he implementation of this baseline relies on this numerical stability, as implemented in Listings 2.1, 2.2, 2.3, and 2.4.

3.1.1 Magnitude error vs frequency

In terms of magnitude error, both M-SDFT and R-SDFT exhibit a minimal discrepancy relative to the standard FFT. Conversely, the approximations introduced in the DC-focused versions yield a significantly larger error. Because this error metric evaluates solely the magnitude of the computed elements, the DC versions remain viable for applications where phase information is non-critical—such as the transmission of constant sinusoidal signals. The magnitude error results are illustrated in Fig. 3.1.

analysis ??? $r < 1$

3.1.2 Complex error (phase) vs frequency

Figure 3.2 presents the complex error analysis across all algorithms. The results confirm the stability considerations discussed previously: the M-SDFT is significantly more stable

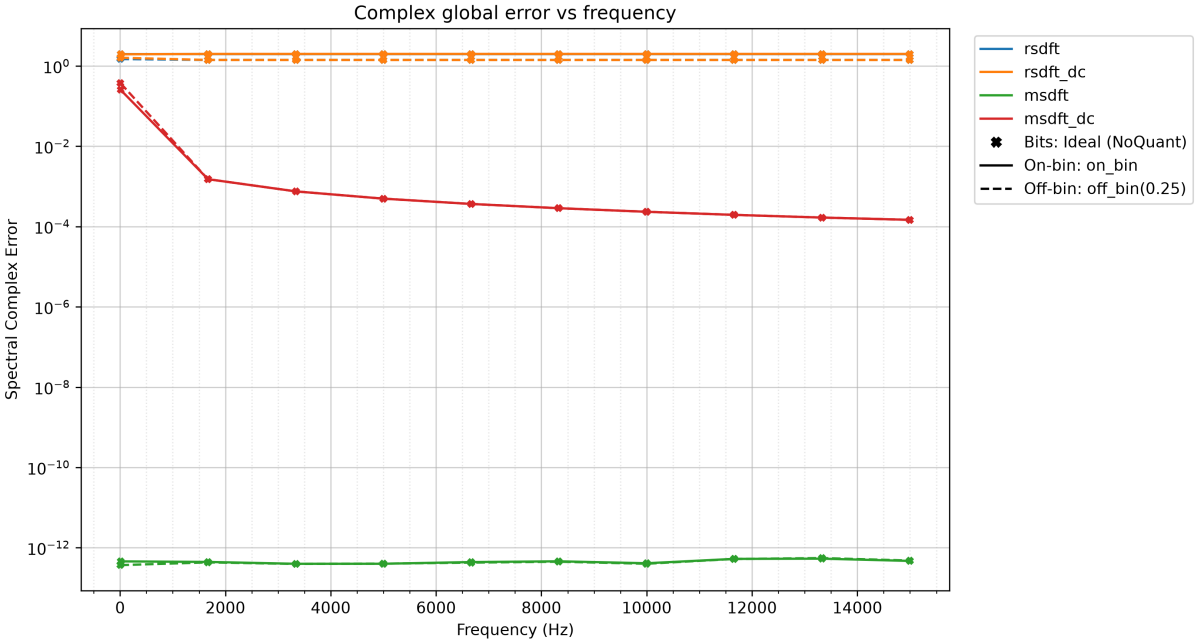


Figure 3.2: Ideal results case analysis, complex error vs frequency, on and off bin cases. $r = 1$ for every algorithm

than the R-SDFT due to its pole placement. Even under high-precision float64 conditions, any slight magnitude deviation in the twiddle factor greater than unity induces a cumulative recursive error in the R-SDFT, resulting in a substantially larger error compared to the M-SDFT. For the DC-based algorithms, the error is higher than the standard sliding formulations due to the simplification of harmonic components.

analysis ??? $r < 1$

3.1.3 Heatmaps

This section evaluates multi-parameter behavior relating amplitude, frequency, and error metrics (both magnitude and complex). As demonstrated in Fig. 3.3 and Fig. 3.4, variations in signal amplitude do not affect the relative comparison under ideal conditions. Instead, the frequency of the input signal influences the error, with R-SDFT exhibiting a higher error than the M-SDFT, and the DC versions showing the largest error.

analysis ??? $r < 1$

3.2 Quantized versions

This section utilizes a custom function, presented in Listing 3.1, to apply quantization to both the input waveforms and the twiddle factors. The primary motivation behind this implementation is the flexibility to evaluate non-conventional bit-widths (e.g., using 5 bits instead of standard 8-bit configurations).

```
1  def quantizer(signal, num_bits:int=8, max_amplitude:float=1.0):
2      if num_bits == "no quant":
3          raise Exception("Quantization is disabled. Please provide a valid number of
               bits.")
4
5      def q(s):
```

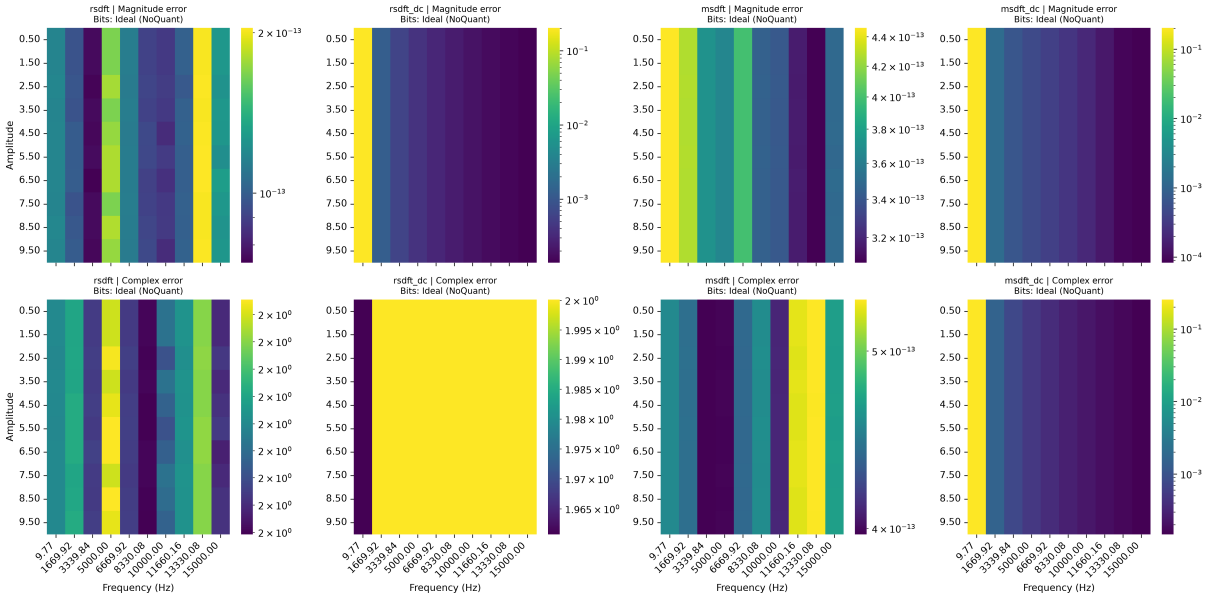


Figure 3.3: Heatmap on-bin (amplitude vs frequency vs errors)

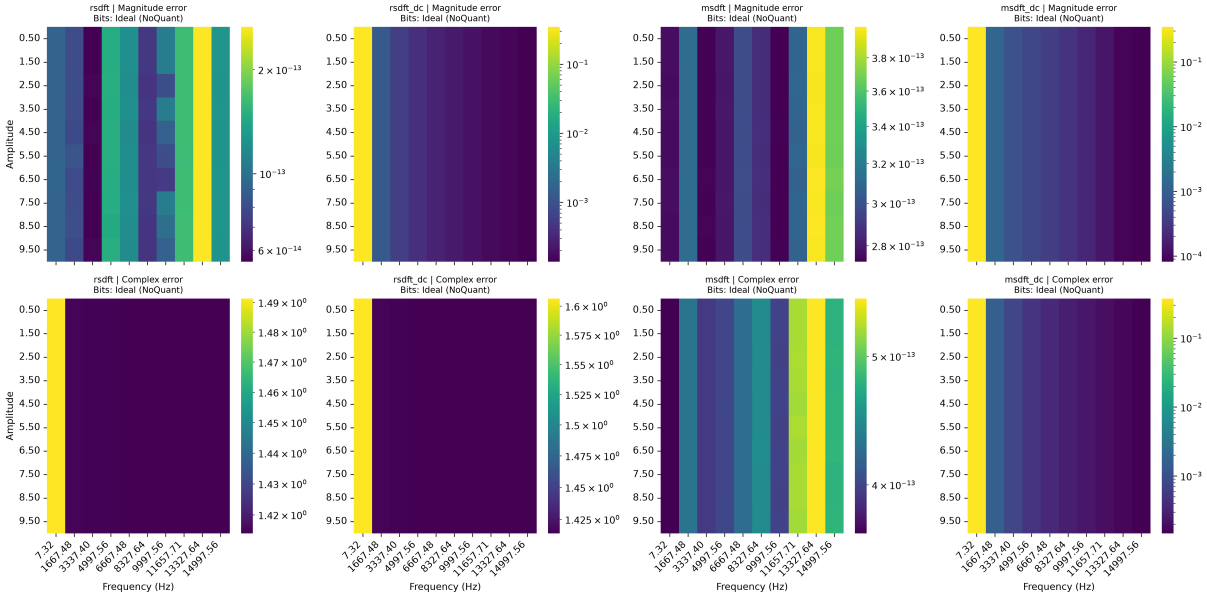


Figure 3.4: Heatmap off-bin (0.25) (amplitude vs frequency vs errors)

```
        half_levels = 2 ** (num_bits - 1)
        max_bits = half_levels      # max 128 with 8 bits
        min_bits = -half_levels + 1 # min -127 with 8 bits

        num = max_amplitude * 2
        denum = half_levels * 2
        resolution = num / denum

        nums = np.round(s / resolution)
        # nums = np.floor(s / resolution)       # wrong idea
        return nums * resolution

    if np.iscomplexobj(signal):
        return q(np.real(signal)) + 1j * q(np.imag(signal))
    else:
        return q(signal)
```

Listing 3.1: Custom quantization funcion Python code

3.2.1 Magnitude error vs frequency

Figure 3.5 illustrates the magnitude error versus frequency across various quantization levels. The results clearly demonstrate that the R-SDFT algorithm is substantially more sensitive to quantization than the M-SDFT counterpart. Additionally, a specific frequency point appears where the M-SDFT DC version undergoes a numerical collapse; this anomalous behavior warrants further investigation in future work. One idea is to explore the behavior of the algorithm under the quantization effects versus the $f_s/2$ and $f_s/4$. In this case $f_s = 40kHz$ and this is the $1/4$ point.
One potential hypothesis is to examine the algorithm's behavior under quantization effects relative to characteristic frequencies such as $f_s/2$ and $f_s/4$. Given a sampling frequency of $f_s = 40\,\text{kHz}$, this collapse occurs precisely at the $f_s/4$ point ($10\,\text{kHz}$). Similarly, Fig. 3.6 depicts the magnitude error as a function of the bit-width for different quantization cases. As expected, the R-SDFT algorithm consistently exhibits a higher error magnitude compared to the M-SDFT architecture.

3.2.2 Complex error (phase) vs frequency

Fig. 3.7 displays the complex error versus frequency across various quantization levels. In this plot, the R-SDFT error curves are positioned above 10^0. Conversely, the M-SDFT curves are clearly discernible, showing a systematic reduction in error as the number of bits increases. Similar to the magnitude analysis, a specific frequency point induces a collapse in DC version of the M-SDFT, a behavior that requires further investigation in future work. Fig. 3.8 illustrates the complex error as a function of the bit-width for different quantization configurations. Comparing these findings with the previous magnitude error results reveals that the complex error is noticeably more sensitive to quantization than the magnitude error alone.

3.2.3 Heatmaps

TO COMPLETE

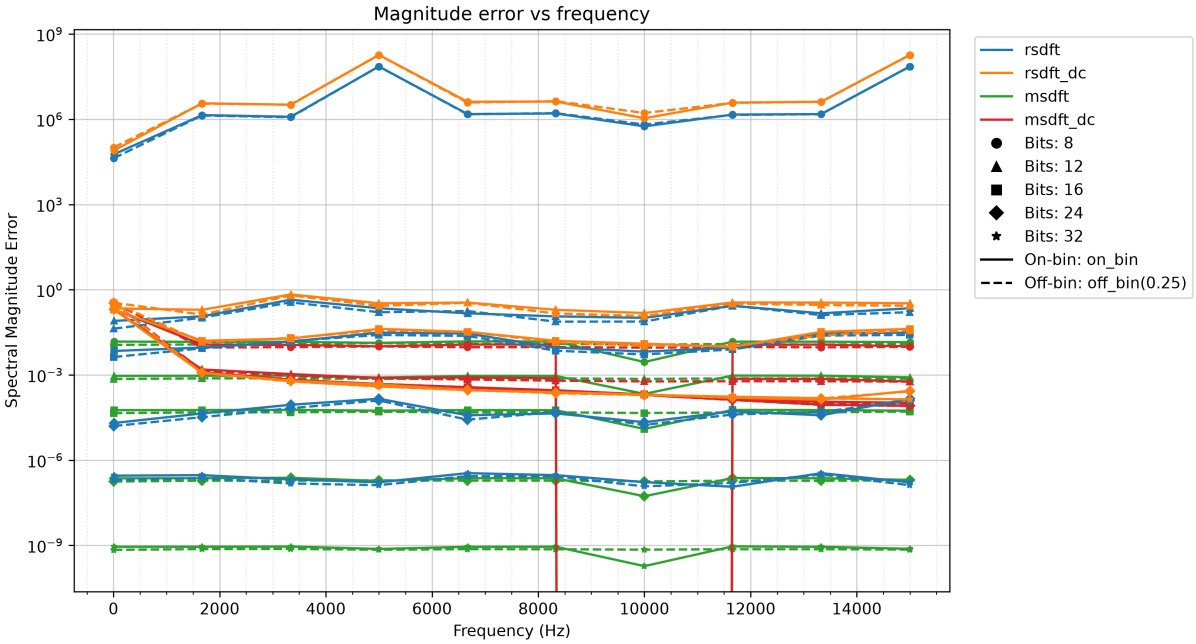


Figure 3.5: Magnitude error vs frequency for different quantization cases. $r = 1$ for every algorithm

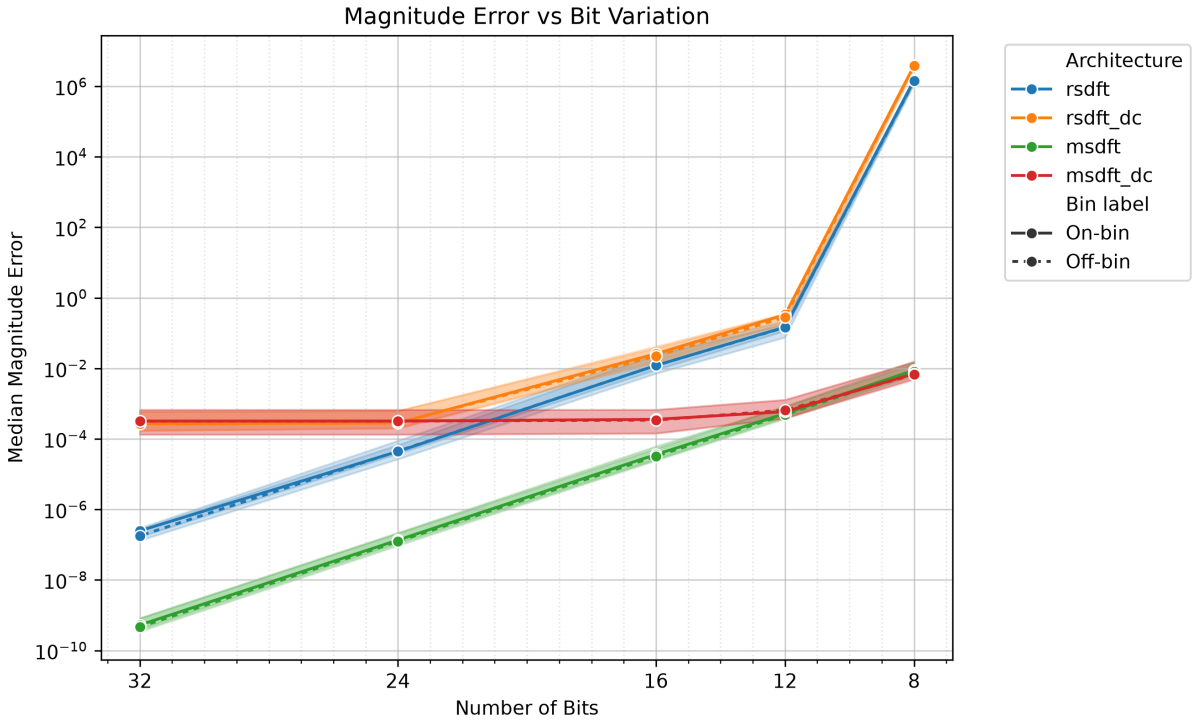


Figure 3.6: Magnitude error vs number of bits for different quantization cases. $r = 1$ for every algorithm

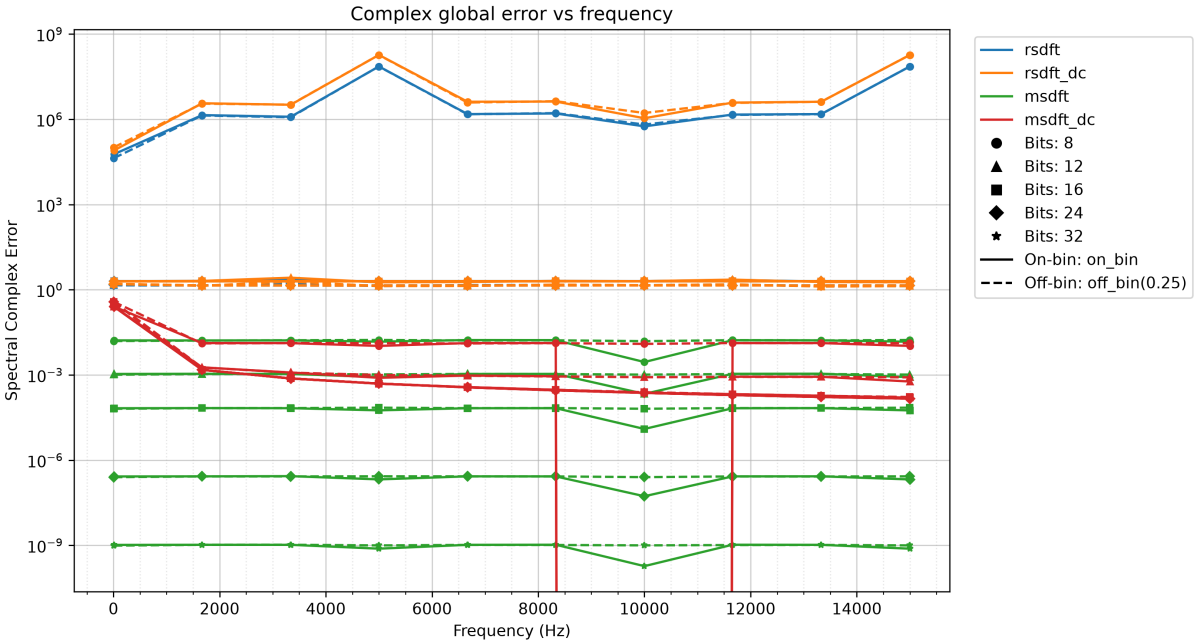


Figure 3.7: Complex error vs frequency for different quantization cases. $r = 1$ for every algorithm

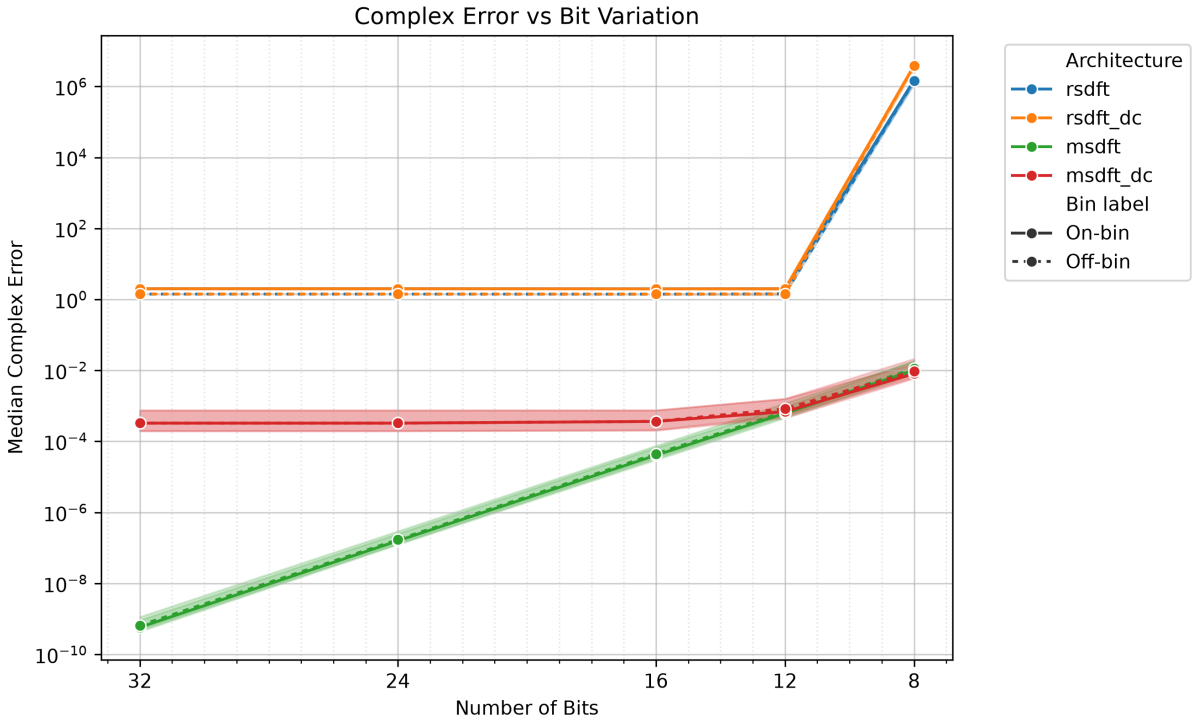


Figure 3.8: Complex error vs number of bits for different quantization cases. $r = 1$ for every algorithm

3.3 Circuital error

This section utilizes a random number generation function to simulate the uncertainty inherent in PCM cells. Based on Fig. 18 of [1], a 6% uncertainty level was selected for a single PCM cell. Consequently, the following boundaries are defined:

```
LOW_ERR = -0.06
HIGH_ERR = 0.06
```

Listing 3.2: PCM cell uncertainty parameters

These parameters are subsequently applied in Listing 3.3 to simulate the hardware uncertainty error:

```
k_vec = np.arange(N)
if hw_err:
    rng = np.random.default_rng()
    k_vec = k_vec + rng.uniform(low=LOW_ERR, high=HIGH_ERR, size=N)
arg = 2 * np.pi * k_vec / N
```

Listing 3.3: PCM cell introduced error

Since the model operates under a normalized amplitude assumption, Listing 3.2 correctly defines 6% of a unit amplitude.

3.3.1 Magnitude error vs frequency

Fig.3.9 presents the magnitude error versus frequency across various quantization levels in the presence of hardware errors. As observed previously, the R-SDFT algorithm remains more sensitive to quantization overall. However, a notable shift occurs here: the M-SDFT algorithm displays a more pronounced error across all frequencies when hardware non-idealities are introduced. Conversely, the R-SDFT error profile appears largely unaffected by the added hardware uncertainty, remaining comparable to the previous non-hardware-error case. Fig.3.10 illustrates the corresponding magnitude error as a function of the number of bits under these hardware-error conditions. As shown in the plot, the R-SDFT algorithm consistently exhibits a pronounced increase in magnitude error as the number of bits decreases.

3.3.2 Complex error (phase) vs frequency

Fig.3.11 presents the complex error versus frequency across various quantization levels in the presence of hardware errors. As observed previously, the error profile is significantly influenced by the introduction of hardware uncertainty for both algorithms. The R-SDFT algorithm continues to exhibit a higher overall error than the M-SDFT, while the latter appears more sensitive to the injected hardware variations. Fig.3.12 illustrates the corresponding complex error as a function of the number of bits, exhibiting behavior similar to the magnitude error versus bit-width analysis: at lower bit resolutions, the R-SDFT algorithm consistently shows a pronounced increase in complex error.

3.3.3 Heatmaps

TO COMPLETE

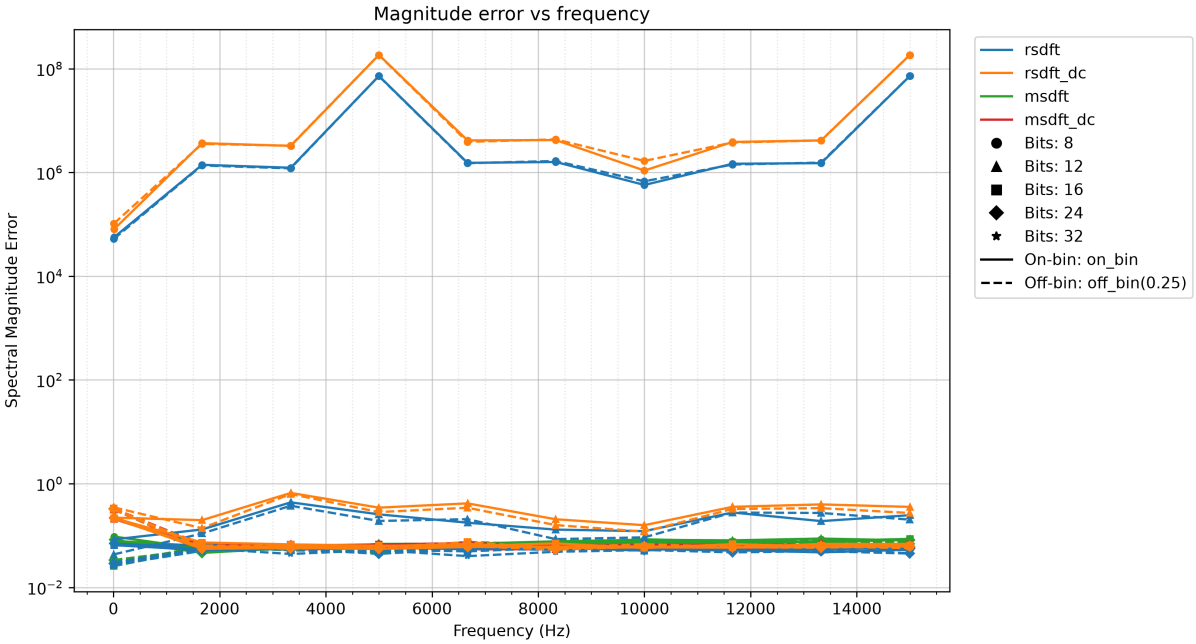


Figure 3.9: Magnitude error vs frequency for different quantization cases and with hardware error. $r = 1$ for every algorithm

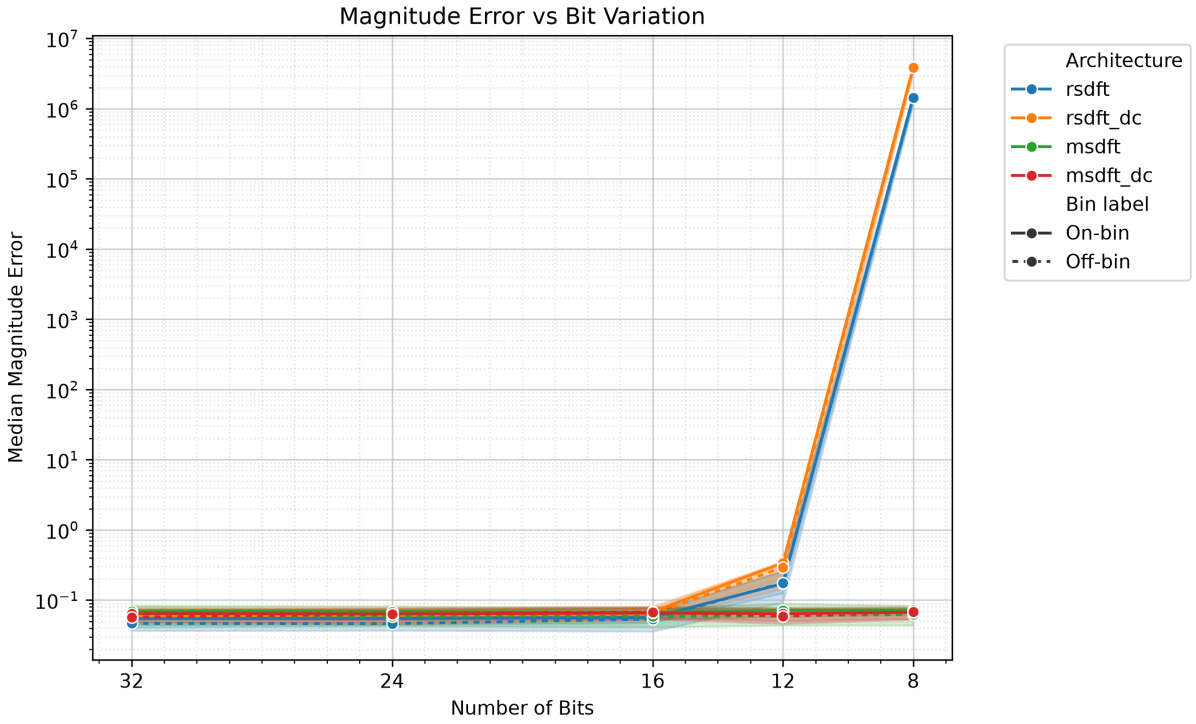


Figure 3.10: Magnitude error vs number of bits for different quantization cases and with hardware error. $r = 1$ for every algorithm

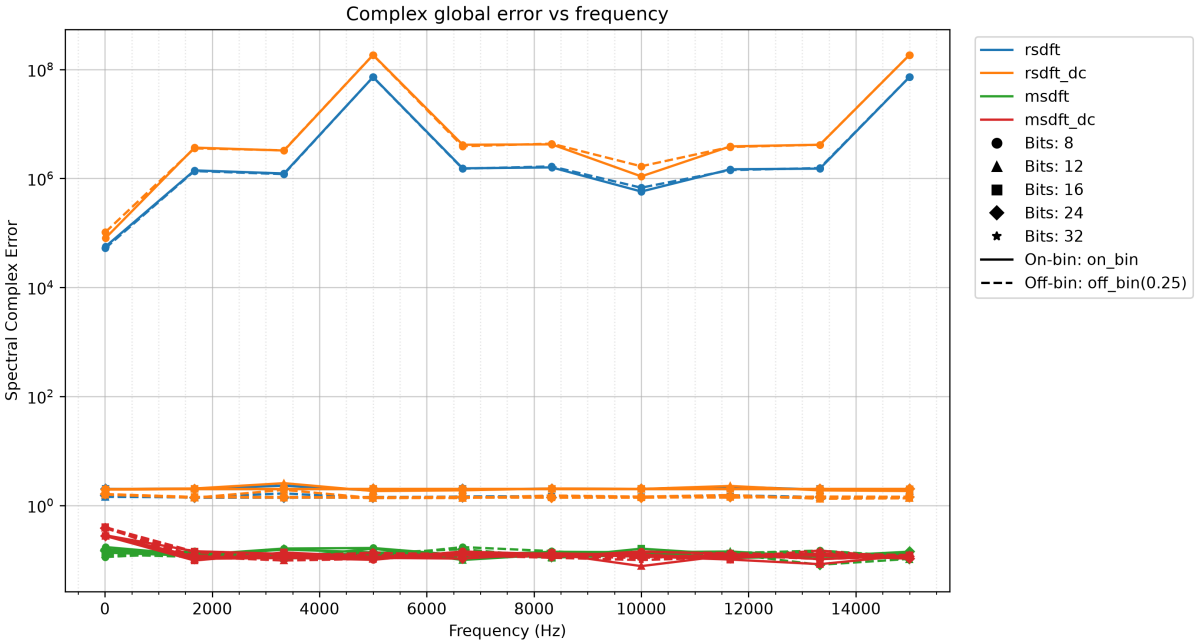


Figure 3.11: Complex error vs frequency for different quantization cases and with hardware error. $r = 1$ for every algorithm

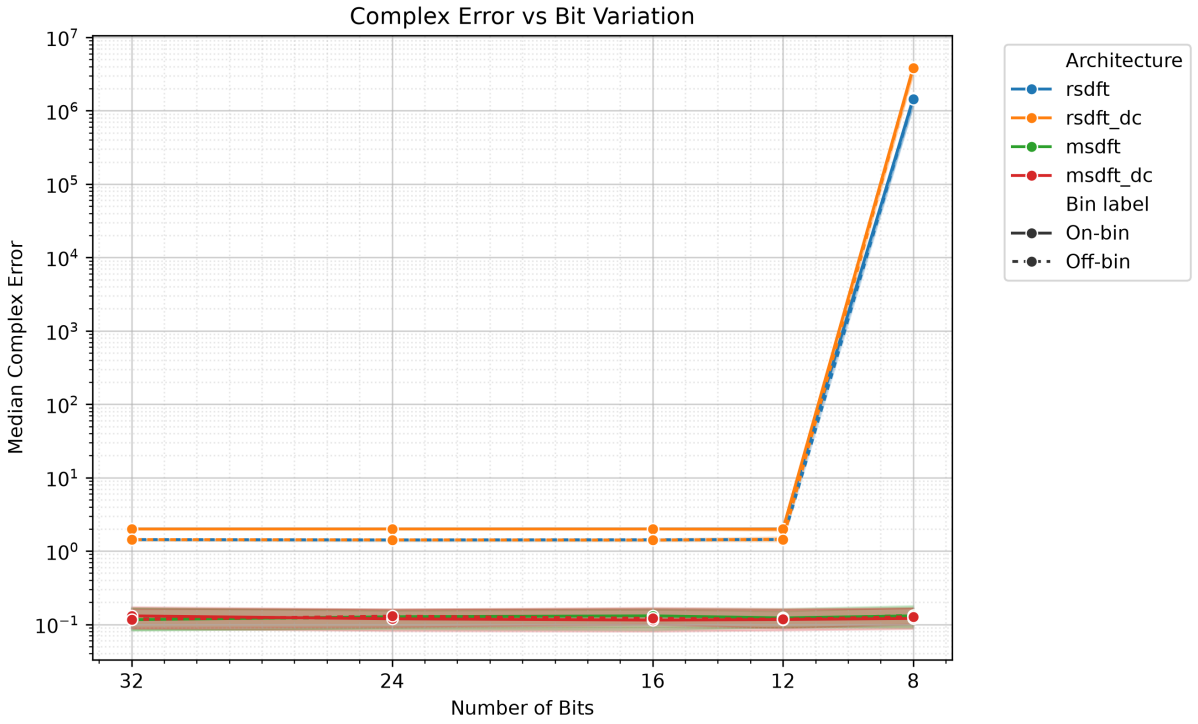


Figure 3.12: Complex error vs number of bits for different quantization cases and with hardware error. $r = 1$ for every algorithm

Chapter 4

Electric circuit idea

This section explores the hardware implementation of a portion of the R-SDFT algorithm. As illustrated in Fig.4.1, the architecture incorporates a digital section where $x[n]$ and $y[n]$ are processed to compute the error $e[n] = x[n] - y[n]$. Each bit of the resulting error controls a switch connected in series with a PCM resistor. When a switch is closed, the corresponding resistor is pulled to ground, allowing current to flow through it. By grouping multiple resistors to a common node, a weighted sum of currents is realized via physical device conductances. This total current is then combined with the currents from other branches, each corresponding to a distinct frequency bin of the DFT. Finally, the aggregated current is converted into a digital quantity and scaled by the number of bins (refer to Fig. 2.3).

4.1 Simulations

In this section is presented the simulations of a small prototype of the circuit idea. In this implementation are used 3 bits PCM resistors. In Fig.4.2 is showed the simulated prototype. The input signal is an hypotetical error from the input-recostruction subtraction signals. The total current is then measured showing the node Kirchhoff's law used in this MAC operation.

As is described in the R-SDFT section the time dependent formnula is built summing two elements: $e[n]$ and the previous spectrum coefficient $X_k[n-1]$ and both are multiplied by the same twiddle factor.

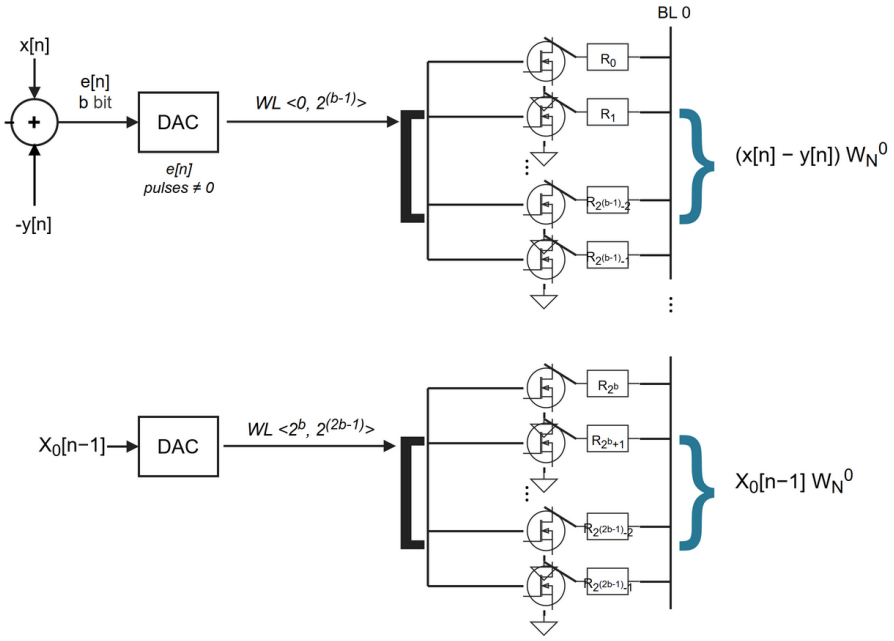


Figure 4.1: Circuit idea for one spectrum coefficient calculated by R-SDFT algorithm

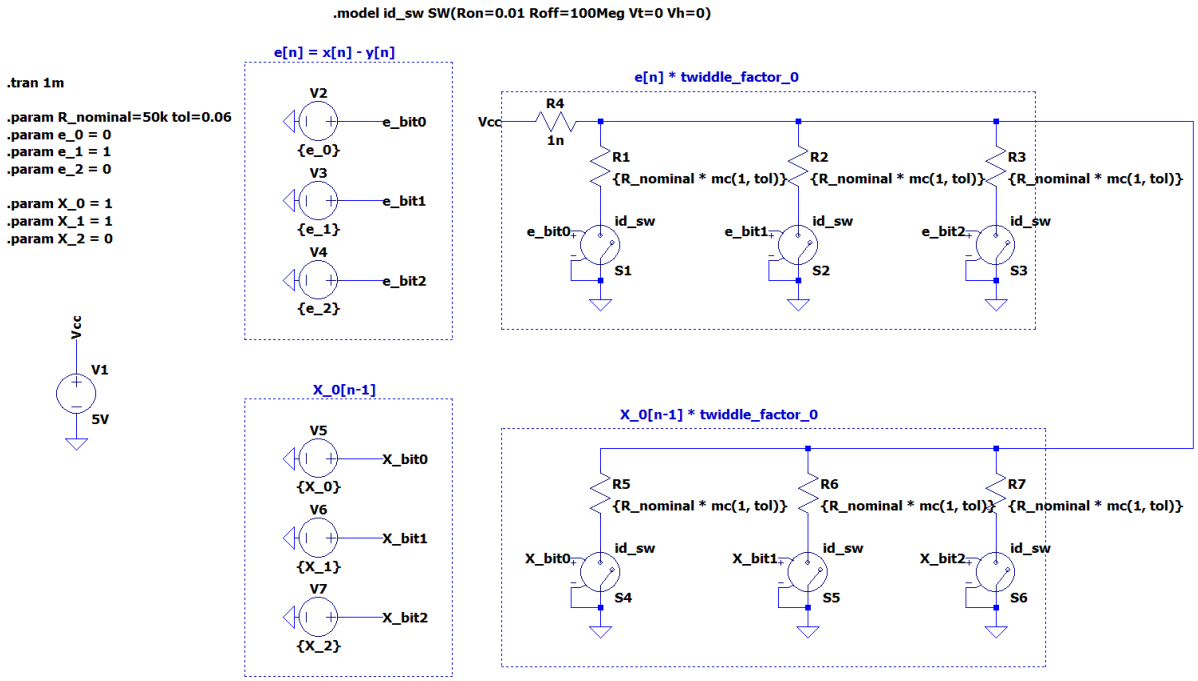


Figure 4.2: Circuit prototype implemented in LTspice. The circuit simulates a 3 bit structure

Chapter 5

Conclusions

Applications where the complex error is not important:
1. Hearingg aids, the human ear needs more the magnitude than the phase of a voice or a sound signal. TO CHECK
2. Constant waves, for example in a telegraph system or a 0/1 signal
3. Power measurement, the power is calculated with the magnitude of the signal.
Applications where the complex error is important:
1. Radar, the phase of the signal is important to understand the distance, the direction and the velocity of an object.
2. Signal reconstruction 3. Filter analyusis, the phase of the signal is important to understand the behavior of a filter.

Bibliography

[1] Alessio Antolini, Eleonora Franchi Scarselli, Antonio Gnudi, Marcella Carissimi, Marco Pasotti, Paolo Romele, and Roberto Canegallo. Characterization and programming algorithm of phase change memory cells for analog in-memory computing. *Materials*, 14(7):1624, 2021.

[2] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, September 2020.

[3] MODIFY. Article title. *JOURNAL NAME*, 322(10):891–921, 1905.